INCH-POUND

MIL-C-70943(AR)

30 July 1990

MILITARY SPECIFICATION

CIRCUIT CARD ASSEMBLY, MICROPROCESSOR: 12922117

This specification is approved for use within the U.S. Army Armament, Munitions and Chemical Command and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification established the requirements for manufacture and acceptance of the Microprocessor circuit card assembly, used in the M90 Radar Chronograph (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards and handbooks. The following specifications, standards and handbooks form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

MILITARY

MIL-F-13926	- Fire Control Materiel, Manufacture and Inspection, General Specification for
MIL-P-14232	- Parts, Equipment and Tools for Army Materiel
MIL-C-24308	- Connectors, Electric, Rectangular, Miniature Polarized Shell, Rack and Panel, General Specification for
MIL-I-45607	- Inspection Equipment, Acquisition, Maintenance and Disposition of

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document, should be addressed to: Commander, U.S. Army ARDEC, ATTN: SMCAR-BAC-S, Picatinny Arsenal, New Jersey 07806-5000 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 1290

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STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	- Quality Assurance Terms and Definitions
MIL-STD-454	- Standard General Requirements for Electronic Equipment
MIL-STD-2164	- Environmental stress screwing process for electronic equipment.

(Unless otherwise indicated, copies of federal and military specifications, standards and handbooks are available from: Standardization Documents Order Desk, Bldg. 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings and publications. The following other Government documents, drawings and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

DRAWINGS (See 6.4)

U.S. ARMY ARMAMENT RESEARCH, DEVELOPMENT AND ENGINEERING CENTER (ARDEC).

12922101	- Microprocessor, Schematic Diagram.
12922105	- EMI Test Procedure
12922117	- Circuit Card Assembly, Microprocessor.
12922145	- M90 Software
12922225	- Environmental Test Methods

SPECIAL PACKAGING INSTRUCTIONS

P12922117	- Special Packaging Instructions for the Circuit Card Assembly, Microprocessor.

(Copies of drawings, publications, and other Government documents required by the contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

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3. REQUIREMENTS

3.1 General description. The Microprocessor circuit card assembly (CCA) of M90 Radar Chronograph is located in the data unit. The CCA processes doppler signal inputs and calculates projectile velocity.

3.2 Components and assemblies. The components and assemblies shall comply with all requirements specified on Drawing 12922117, all associated drawings, and with all requirements specified in applicable specifications and standards. Where applicable, the CCA shall meet the requirements of MIL-F-13926 and MIL-STD-454 as follows:

a. Dimensions and tolerances
b. Inorganic protective surface finishings
c. Part identification and marking
d. Workmanship
e. Soldering electrical connections

3.3 Performance requirements.

General description. The Microprocessor circuit card assembly is composed of the following 4 basic functional blocks:

a. Microprocessor circuitry
b. Filter circuitry
c. Measuring circuitry
d. Voltage Monitor and Switches circuitry

The Microprocessor circuitry is centered around the INTEL 8085 microprocessor. The Microprocessor controls the M90 Radar Chronograph using Input and Output ports to collect information and set system mode. The Microprocessor polls via input ports, all of the front panel control switches except the power switch, and the microprocessor reads this information from the port. The Microprocessor reads the Doppler frequency from a programmable counter which counts the filtered doppler signal. The computed muzzle velocity data is sent to the display card via the BCD DATA PORT and the DIGIT SELECT PORT. Projectile velocity data is transmitted and control signals are received, (by the 8251 USART) through the remote control connector.

3.3.1 CCA logic. The CCA shall be capable of measuring a Doppler Signal Frequency and convert it to velocity representation. The CCA shall be capable of driving a display unit with the measurement results:

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3.3.1.1 Logic states:

a. STAND-BY - when in STAND BY state the CCA shall be able to recall data.

b. READY - when in READY state the CCA shall sample input commands and perform measurement.

3.3.1.2 Measurement algorithm:

a. The CCA shall be manually set to the READY state.

b. The CCA shall use a trigger mechanism in order to detect the presence of a Doppler signal at the CCA input. The moment of arrival shall be defined as TO.

c. Starting at TO the CCA shall count N input signal cycles. 4 values for N shall be manually controlled by setting CALIBER. Concurrently, the CCA shall measure the elapsed time T required to count the N cycles.

d. After completing the count of N cycles, the CCA shall perform 7 measurements of the elapsed time, T1-T7, required to count 256 input signal cycles. The time delay between each measurement shall be T/4.

e. The CCA shall used the least square method to develop FO, the input signal frequency at TO.

f. The CCA shall translate FO Doppler frequency to velocity using the following formula:

VO = .014242 x FO [m/s]

g. The CCA shall update the Display Driver with the calculated VO.

h. The CCA shall enter the STAND BY state.

3.3.2 Connector. The CCA shall physically interface to the M90 Radar Chronograph via a D-type, 78 pin connector, type M24308/4-15 in accordance with MIL-C-24308. The connector is designated as P5. Pin reference designation shall be in accordance with TABLE I - Microprocessor circuit Card Assembly - Connector Contacts.

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TABLE I. Microprocessor circuit card assembly connector contacts

Connector	Pin	Signal Name
P5	1	SPARE
P5	2	DISP 2
P5	3	/DIGIT SEL 4
P5	4	V RANGE 03
P5	5	V RANGE 01
P5	6	DISP SW 01
P5	7	DISP SW 02
P5	8	/DIGIT SEL 5
P5	9	SPARE
P5	10	CALIBER 00
P5	11	/B EN 256 CNTR
P5	12	/BUFF REF DOPP
P5	13	DISP 4
P5	14	/DIGIT SEL 1
P5	15	/DIGIT SEL 3
P5	16	SPARE
P5	17	SER DATA IN
P5	18	DTR
P5	19	SER DATA OUT
P5	20	SPARE
P5	21	+5V
P5	22	+5V
P5	23	GND
P5	24	GND
P5	25	GND
P5	26	-12V
P5	27	-5V
P5	28	REF DOPP
P5	29	/PBF NO
P5	30	DOPPLER IN(-)
P5	31	SQ DOPP
P5	32	/STANDBY
P5	33	DOPP SIG
P5	34	FILT DOPP
P5	35	SPARE
P5	36	DISP 1
P5	37	BLANK DIGIT
P5	38	V RANGE 02

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Connector	Pin	Signal Name
P5	39	DISP SW 00
P5	40	V RANGE 00
P5	41	DISP SW 03
P5	42	SPARE
P5	43	CALIBER 01
P5	44	/CPU RESET
P5	45	T-T/4 CNTR OUT
P5	46	3.6M CTR CLK
P5	47	DISP 8
P5	48	CTS
P5	49	DSR
P5	50	/DIGIT SEL 2
P5	51	DIGT SEG G
P5	52	/DIGIT SEL 0
P5	53	RTS
P5	54	SER SIG GND
P5	55	DATA SET RDY
P5	56	SPARE
P5	57	+5V
P5	58	+5V
P5	59	+12V
P5	60	GND
P5	61	GND
P5	62	-12V
P5	63	-5V
P5	64	/TRIG
P5	65	/PBF NC
P5	66	L3.6M CNTR OUT
P5	67	DOPPLER IN(+)
P5	68	+24V INPUT PWR MON
P5	69	READY
P5	70	LOW PASS OUT
P5	71	RELAY POWER
P5	72	SPARE
P5	73	SPARE
P5	74	+12V
P5	75	+12V
P5	76	GND
P5	77	GND
P5	78	-12V

3.3.3. Display driver. The CCA shall be capable of driving a display unit with 4 digits and one digit to the nearest tenth (.1). The CCA shall drive a hexadecimal display element to display a positive error indication or the number of measurement rounds in the average displaying mode. The CCA shall drive a display unit with the following requirements:

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a. Number of Display Elements (DE): 6.

b. Data: 4 bit binary coded, Common to all Display Elements.

c. Each Display Element shall be separately latched with the data to be displayed. The hexadecimal Display Element shall have a blanking command input (1 bit).

d. P5 pin designation shall be in accordance with TABLE II.

TABLE II - Display driver requirements

Pin of P5	PIN DESIGNATION	FUNCTION	LOGIC
36	DISP 1	BIT 0 OF DATA (LSB)	POSITIVE
2	DISP 2	BIT 1 OF DATA	POSITIVE
13	DISP 4	BIT 2 OF DATA	POSITIVE
47	DISP 8	BIT 3 OF DATA (MSB)	POSITIVE
52	DIGIT SEL 0	LATCH ENABLE DE #1	EDGE TRIG.
14	DIGIT SEL 1	LATCH ENABLE DE #2	EDGE TRIG.
50	DIGIT SEL 2	LATCH ENABLE DE #3	EDGE TRIG.
15	DIGIT SEL 3	LATCH ENABLE DE #4	EDGE TRIG.
3	DIGIT SEL 4	LATCH ENABLE DE #5	EDGE TRIG.
8	DIGIT SEL 5	LATCH ENABLE DE #6	EDGE TRIG.
37	DIGIT BLANK	BLANKING DE #1	LOW = NOT BLANK

3.3.4 DISP SW. The CCA shall be capable of automatically storing and recalling the data for the last eight rounds measured on demand by controlling the DISP SW inputs. It will also calculate the average and the number of rounds used in the average. Averaging shall commence with the determination of data of the second round. The CCA shall have 4 command inputs designated as:

DISP SW 00: PIN 39 OF P5
DISP SW 01: PIN 6 OF P5
DISP SW 02: PIN 7 OF P5
DISP SW 03: PIN 41 OF P5.

The CCA shall recall data of the last rounds or the average and number of rounds that the average was calculated for in accordance with TABLE III.

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TABLE III - DISP SW inputs

Pins of P5				Measurement Rounds
41	7	6	39	
1	1	1	1	Current
1	1	1	0	Current - 1
1	1	0	1	Current - 2
1	1	0	0	Current - 3
1	0	1	1	Current - 4
1	0	1	0	Current - 5
1	0	0	1	Current - 6
1	0	0	0	Current - 7
0	1	1	1	Average, No. of rounds

0 - TTL LOGIC LOW, 1 - TTL LOGIC HIGH

3.3.5 Accuracy. The CCA shall determine the velocity VO with a precision of .25 percent or .5 meters per second (whichever is greater) in the velocity range of 150 to 1200 meters per second.

3.3.6 Self test. The CCA shall incorporate a self test to check functionally by simulating a Doppler signal at CCA input. The CCA shall simulate a Doppler signal of 64 KHz. In this state the CCA shall ignore a real Doppler input. Self test is resumed by setting pins 40, 5, 38, 4 of P5 to the TTL logic combination of 1,1,1,1.

3.3.7 Software. Logic operation of the CCA shall be controlled by software as defined by Drawing 12922145 and Associated Documents (software product specification).

3.3.8 V Range. The CCA shall have a measurement range of 150 to 1250 meters per second divided into 10 sub-ranges as follows:

Range	Velocity (m/s)
1	150 - 300
2	300 - 400
3	400 - 500
4	500 - 600
5	600 - 700
6	700 - 800
7	800 - 900
8	900 - 1000
9	1000 - 1100
10	1100 - 1250

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The CCA shall have 4 command inputs designated as:

V Range 00: PIN 40 of P5
V Range 01: PIN 5 of P5
V Range 02: PIN 38 of P5
V Range 03: PIN 4 of P5

Measurement range shall be selected by applying logic levels to V Range input pins of P5 in accordance with TABLE IV. The CCA shall limit the input Bandwidth by using a variable bandpass filter. -3DB frequency limits and insertion loss shall be in accordance with TABLE IV.

TABLE IV - Measurement ranges

Pins of P5				Range		-3DB frequencies		Minimum Gain at optimum pass frequency
4	38	5	40	Min	Max	Low	High	
1	1	1	1	900	1000	58+/-5KHz	74+/-5KHz	.20
1	1	1	0	150	300	10+/-2KHz	25+/-2KHz	.45
1	1	0	1	300	400	17+/-2KHz	32+/-2KHz	.45
1	1	0	0	400	500	24+/-2KHz	40+/-2KHz	.35
1	0	1	1	500	600	31+/-2KHz	45+/-2KHz	.30
1	0	1	0	600	700	36+/-2KHz	54+/-2KHz	.25
1	0	0	1	700	800	44+/-2KHz	62+/-5KHz	.22
1	0	0	0	800	900	49+/-5KHz	70+/-5KHz	.20
0	1	1	1	900	1000	58+/-5KHz	74+/-5KHz	.10
0	1	1	0	1000	1100	63+/-5KHz	81+/-5KHz	.10
0	1	0	1	1100	1250	69+/-5KHz	89+/-5KHz	.10
0	1	0	0	300	400	17+/-2KHz	32+/-2KHz	.45

0 - TTL LOGIC LEVEL LOW, 1 - TTL LOGIC LEVEL HIGH

3.3.9 Automatic reset. When power is applied, the CCA shall resume restart procedure and go to STAND-BY state.

3.3.10 Press before fire (PBF). Command input to the CCA to toggle between READY and STANDBY states, designated as PBF NO, Pin 29 of P5 and PBF NC, Pin 65 of P5. Normal conditions are for pin 29 to be at TTL logic level high and Pin 65 at TTL Logic level low. When pin 29 goes to logic level low and pin 65 to logic level high, the CCA shall toggle from Ready to Stand By state, and vice-versa.

3.3.11 CPU reset. External reset input to the CCA microprocessor. The reset input shall be in Pin 44 of P5, designated as CPU RESET. The signal input shall be active low, TTL logic levels.

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3.3.12 Indicator lamps. The CCA shall provide means to activate two indicator lamps of 28 volts, 50 mA. Each indicator lamp shall have one pole connected to 28V and the other pole connected to the CCA pin 32 of P5 designated as /STANDBY and pin 69 of P5 designated as /READY. /STANDBY indicator shall be activated when the CCA is in STAND BY state. /READY indicator shall be activated when the CCA is in READY State.

3.3.13 Antenna control. The CCA shall drive an on/off control line correlated to READY or STAND-BY states respectively. The CCA shall supply a 12 volt command through a 680 ohm resistor on the command line, Pin 67 of P5. When in READY state, the Return command line, Pin 30 of P5 shall be grounded through a 680 ohm resistor to CCA ground. When in STAND BY state, the Return command line shall be isolated from CCA ground.

3.3.14 +24V INPUT PWR MON. The CCA shall have the means of sensing a voltage of 30 V maximum and detect a level of under 16.4 +/- .5 V. The input voltage shall be applied to the CCA on pin 68 of P5, designated as +24V INPUT PWR MON. When the +24V INPUT PWR MON voltage goes under 16.4 +/- .5 V, Indicator Lamps and Antenna Control shall be disabled.

3.3.15 Caliber. The CCA shall accept manual control of the value N used in the measurement algorithm to be one of the following:

N1 = 2107
N2 = 4214
N3 = 8444
N4 = 12643

The CCA shall have 2 command inputs designated as CALIBER 00, Pin 10 of P5 and CALIBER 01, Pin 43 of P5.

The CCA shall set the measurement constant N in accordance with TABLE V.

TABLE V - N values vs caliber

Pins of P5		Value of N
43	10	
1	1	2,107
1	0	4,214
0	1	8,444
0	0	12,643

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0 - TTL Logic Level Low 1 - TTL Logic Level High

3.3.16 Serial communication. The CCA shall be capable of transmitting and receiving serial data. The CCA shall have serial communication input/output lines in accordance with TABLE VI. Voltage levels shall be in accordance with the requirements of RS-232.

TABLE VI - Serial communication lines

Pin of P5	REF DESIGNATION	FUNCTION	ACTIVE
19	SER DATA OUT	Serial data output	YES
53	RTS	Transmitter hand-shake	NO
18	DTR	Receiver hand-shake	NO
17	SER DATA IN	Serial data in	YES
48	CTS	Transmitter hand-shake	NO
54	SER SIG GND	GROUND	
49	DSR	Receiver hand-shake	NO
55	DATA SET READY	+12V with 2KOHM Resistor	

3.3.17 Test signals. The CCA shall provide the following signals in order to enable monitoring of CCA functions:

a. DOPP SIG. - The input doppler signal to the CCA following input circuitry, PIN 33 of P5.

b. LOW PASS OUT - Doppler signal following a low pass filter, Pin 70 of P5.

c. FILT D0PP - Doppler signal after a bandpass filter, Pin 34 of P5.

d. SQ DOPP - Zero crossing detection of the Doppler signal after the band pass filter, Pin 31 of P5.

e. REF DOPP - output of a comparator which compares the filtered doppler signal with 9 db level above the noise average before the moment TO, Pin 28 of P5.

f. TRIG - output of trigger mechanism, Pin 64 of P5. Negative TTL Logic signal that indicates that the filtered Doppler signal has passed an internal preset level. The minimum Doppler

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signal that will reenable the CCA trigger circuitry shall be not less than 6 volt peak to peak.

g. 3.6M CTR CLK - 7 groups of 256 Doppler signal cycles output while the CCA performs a measurement, Pin 46 of P5.

h. /B EN 256 CNTR - 7 pulses output while the CCA performs a measurement, Pin 11 of P5.

i. /L 3.6M CNTR - 7 pulses output while the CCA performs a measurement, Pin 66 of P5.

j. /BUF REF DOPP - REF DOPP after buffering, Pin 12 of P5.

k. /T-T/4 CNTR OUT - output pulses at the rate of 4/T pulses per second, Pin 45 of P5.

3.3.18 Processor element. The CCA shall be devised around the INTEL 8085 processor.

3.3.19 Power requirements. Power requirements of the CCA shall not exceed the limits of TABLE VII.

TABLE VII - Power requirements

P5-PIN NO.	DESIGNATION	VOLTAGE	CURRENT (MAX)
21,22,57,58	+5V	+5 +/- .25 V	1.2 A
27,63	-5V	-5 +/- .5 V	40 mA
59,74,75	+12V	+12 +/- .5 V	250 mA
26,62,78	-12V	-12 +/- .5 V	200 mA
23,24,25,60, 61,76,77	GND	VOLTAGE GROUND	

3.3.20 DOPPLER IN. The CCA shall have a differential Doppler Signal input AC coupled that can withstand the following conditions:

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3.3.20.1 Voltage level.

P5 PIN NO.	DESIGNATION	AC VOLTAGE MIN	AC VOLTAGE MAX	DC VOLTAGE (MAX)
67	DOPPLER IN (+)	-10v	+10v	+12V
30	DOPPLER IN (-)	-10v	+10V	

3.3.21 Relay power. The CCA shall have a 24 +/- 6 Volt input at pin 71 of P5 for relay power.

3.4 Environmental requirements.

3.4.1 High temperature. The CCA shall meet the requirements of 3.2 and 3.3 when thermally stabilized at +125 degrees F. The CCA when subjected to storage temperature of +150 degrees F shall meet the requirements of 3.2 and 3.3 when returned and thermally stabilized at ambient temperature.

3.4.2 Low temperature. The CCA shall meet the requirements of 3.2 and 3.3 when thermally stabilized at -50 degrees F when returned and thermally stabilized at ambient temperature.

3.4.3 Conformal coating. Each CCA, after completion of production state and before ESS tests, shall be conformally coated using AR type coating, 0.003 +/-0.002 inches thickness.

3.4.4 Environmental stress screening (ESS) temperature. The Microprocessor circuit shall meet the requirements in 3.2 and 3.3, after being subjected to 40 hours of temperature cycling between 0 degrees F and +125 degrees F in accordance with 5.1.2.1 of MIL-STD-2164.

3.4.5 Environmental stress screening (ESS) vibration. Microprocessor circuit card shall meet the requirements in 3.2 and 3.3, after being subjected to random motion in the range of 20 to 2000 Hz for 5 minutes at an axis perpendicular to the largest surfaces of the CCA. Acceleration spectral density is 6.06 g. RMS in accordance with 5.5.1 of MIL-STD-2164.

3.4.6 EMI. The CCA shall meet the requirements of drawing 12922105 when integrated into an M90 RADAR CHRONOGRAPH (NSN: 1290-01-073-0764) as Government Furnished Material.

3.5 First article inspection. When specified in the contract or purchase order, a sample shall be subjected to first article

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inspection in accordance with the technical provisions herein (see 4.3).

3.6 Workmanship. The workmanship requirements of MIL-F-13926 and MIL-STD-454 shall apply.

3.7 Fabrication. The CCA shall be manufactured in accordance with Drawings 12922101 and 12922117.

3.8 Materials. Materials shall be in accordance with applicable drawings and specifications.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Reference shall be made to MIL-STD-109 to define quality assurance terms used herein.

4.1.1 Responsibility for compliance. All items meet all requirements of sections 3 and 5 and all applicable drawings. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 General provisions. The component and subassembly inspection requirements of MIL-F-13926 and MIL-STD-454 form a part of the Quality Assurance Provisions of this specification.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).
c. Special sampling (see 4.5).

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4.2.1 Inspection conditions. Unless otherwise specified, all examinations and tests shall be conducted at the following standard ambient conditions:

a.	Temperature	+60 degrees to +90 degrees F.
b.	Relative humidity	90% RH maximum
c.	Pressure	28 to 32 inches of Hg.

4.3 First article inspection.

4.3.1 First article submission. The contractor shall submit a first article sample as designated by the contracting officer consisting of three CCA for inspection in accordance with 4.3.2.

4.3.2 Inspections to be performed. First article assemblies, components and test specimens may be subjected by the Government to any or all of the examinations and tests specified in TABLE II and TABLE III herein, and to any or all requirements of the applicable drawings.

4.3.3 Rejection. If any assembly, component or test specimen fails to comply with any of the applicable requirements, the first article sample shall be rejected. The Government reserves the right to terminate its inspection upon any failure of any CCA to comply with any of the stated requirements.

4.4 Quality conformance inspection.

4.4.1 Inspection lot formation. The term "inspection lot" is defined as a homogeneous collection of units of product from which a representative sample is drawn or which is inspected 100 percent to determine conformance with applicable requirements. Units of product selected for inspection shall represent only the inspection lot from which they are drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. Homogeneity shall be considered to exist provided the inspection lot has been produced by one manufacturer, in one unchanged process, using the same materials and methods, in accordance with the same drawings, same drawing revisions, same specifications and same specification revisions, and complies with the provisions for submission of product as specified in MIL-STD-105. All material submitted for inspection in accordance with this specification shall comply with the homogeneity criteria specified herein, regardless of the type of inspection procedure which is being applied to determine conformance with requirements.

4.4.2 Examinations and tests.

4.4.2.1 Components, assemblies and subassemblies. All components, assemblies and subassemblies shall be inspected in

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accordance with the inspection provisions contained in this specification and Quality Assurance Provisions (QAP) listed in the Technical Data Package (TDP). In the absence of QAPs, the applicable quality assurance provisions of MIL-F-13926 shall apply.

4.4.2.2 Final acceptance inspection. Subsequent to first article approval, examination and tests related to Section 3 herein shall be performed in accordance with the sampling plans specified in 4.4.2.3 herein. Examination and test for packaging and marking shall be in accordance with MIL-P-14232 and Section 5 herein. The tabulated classification of defects in 4.4.2.3 shall constitute the minimum inspection to be performed by the supplier after first article approval and prior to Government acceptance or rejection by item or lot.

4.4.2.3 Functional tests. The tests in TABLE VIII shall be performed as noted under the characteristics.

TABLE VIII. Functional tests
(100% Inspection)

No.	Characteristics	Requirement	Test Procedure
Critical:	None	None	N/A
Major:			
101.	Performance	3.3	4.8
102.	Conformal Coating	3.4.3	Visual
103.	ESS Temperature	3.4.4	4.7.1
104.	ESS Vibration	3.4.5	4.7.2
105.	Workmanship	3.6	Applicable Drawing/ Visual
106.	Fabrication	3.7	Applicable Drawing/ Visual
107.	Materials	3.8	Applicable Drawing/ Visual
Minor: None			

4.5 Special sampling. Three circuit card assemblies shall be selected at random as a special sample from each lot or each 150 produced, whichever occurs first. The sample shall meet the requirements and tests in TABLE IX.

TABLE IX. Environmental sampling tests.

No.	Characteristics	Requirement	Test Procedure
301.	High temperature	3.4.2.1	4.7.3.1
302.	Low temperature	3.4.2.2	4.7.3.2

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No.	Characteristics	Requirement	Test Procedure
303.	EMI	3.4.6	4.7.3.3

4.6 Inspection equipment.

4.6.1 Government-furnished inspection equipment. Where the contract provides for Government-furnished test equipment, supply and maintenance of test equipment shall be in accordance with the applicable requirements of MIL-I-45607.

4.6.2 Contractor-furnished inspection equipment.

4.6.2.1 Government design. Unless otherwise specified in the contract, all inspection equipment identified by drawing number in specifications or QAPs forming a part of the contract shall be supplied by the contractor in accordance with the design specified. The contractor may, however, propose alternatives to Government designs for approval in accordance with 6.4.

4.6.2.2 Contractor design. The contractor shall design and supply inspection equipment compatible with the "Methods of Inspection" specified in 4.7 of this specification and with the component inspection procedures specified in "Examination" and "Test Facilities" requirements of MIL-F-13926 whenever Government designs are not specified (see 4.6.2.1). Since tolerance of test equipment is normally considered to be within 10 percent of the product tolerance for which it is intended, this inherent error in the test equipment design must be considered as part of the prescribed product tolerance limit. Thus, concept, construction, materials, dimensions, and tolerances used in the design of test equipment will reliably indicate acceptability of a product which does not exceed 90 percent of the prescribed tolerance limit and permit positive rejection when non-conforming. Construction shall be such as to facilitate routine calibration of test equipment. Contractor inspection equipment designs shall be submitted for approval in accordance with 6.4.

4.6.2.3 Inspection equipment design requirements.

4.6.2.3.1 Temperature testing. The volume of the test chamber shall conform to the test facilities requirements of MIL-F-13926. The air temperature at the control sensor shall be in accordance with the specified requirements within plus or minus 2.5 degrees F (1.4 degrees C). The temperature gradient across the cross-sectional area occupied by the test item shall not exceed 0.5 degrees F per foot in any direction. The heat source of the test chamber shall be located so that radiant heat, if used, will not fall directly on the test item. Thermocouples or equivalent temperature sensors utilized to determine or control the specified

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chamber temperature shall be centrally located within the test chamber within the supply airstream, or in the return airstream, whichever provides the specified test condition at the item under test and shall be baffled or otherwise protected against radiation effects.

4.6.2.3.2 Fabrication, setup and calibration. Design and fabrication of test equipment shall be such that it incorporates the features of rigidity, stability, and ease of maintenance. Setup and calibration equipment necessary for certification of the testing equipment accuracy shall be furnished and will become a part of the testing equipment. Operating and calibration instruction shall be prepared and submitted to the procuring activity.

4.7 Methods of inspection. Unless otherwise specified in the contract, the following procedures shall be utilized to conduct specified tests. If the contractor chooses to use alternate test procedures, a detailed description of the set-up and procedure shall be submitted to the government procuring agency for approval prior to use.

4.7.1 ESS temperature. Testing equipment utilized in this test shall conform to the requirements of 4.6.2.3.1. The circuit card shall be placed in the chamber and subjected to a 40-hour run-in period which includes vibration and operation at temperature extremes. The temperature shall be raised gradually at a rate not to exceed 2 degrees F per minute until thermally stabilized at +125 degrees F for 4 hours duration. Then the temperature of the chamber shall be decreased to 0 degrees F at the same rate for another 4 hours. Subsequent to the above temperature extremes (+125 degrees F and 0 degrees F), the CCA shall not have been damaged or adversely affected and shall meet the requirements in 3.2 and 3.3 of this specification for the entire 40-hour run-in period.

4.7.2 ESS vibration. The CCA in an operating mode shall be exposed to a 5 minute burst of random vibration. Failures occurring during this five minute test shall be accrued (if possible) and corrected at the conclusion of the 5 minute period in accordance with the requirements of 3.2 and 3.3. The random vibration spectrum shall be:

20 - 80	Hz	at 3 dB/octave rise
80 - 350	Hz	at 0.04 g sq/Hz
350-2000	Hz	at 3 dB/octave rolloff

The motion shall be applied perpendicular to the largest surface of the CCA.

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4.7.3 Environmental sampling tests.

4.7.3.1 High temperature. This test shall be conducted in accordance with Drawing 12922225. Subsequent to the exposure, the CCA shall not be damaged or adversely affected and shall meet the requirements of 3.4.1.

4.7.3.2 Low temperature. This test shall be conducted in accordance with Drawing 12922225. Subsequent to the exposure, the CCA shall not be damaged or adversely affected and shall meet the requirements of 3.4.2.

4.7.3.3 EMI. The test shall be conducted in accordance with Drawing 12922105 when the CCA integrated into an M90 RADAR CHRONOGRAPH.

4.8 Performance. The performance parameters of 3.3.1 thru 3.3.21 shall be demonstrated either manually or using a contractor designed, Government-approved, automatic test fixture simulating the conditions and making the measurements of providing a visual indication of a PASS or FAIL condition. Test shall be conducted before ESS testing, and during and after completion of the ESS testing. The performance test shall verify that each of the following CCA functional characteristics conform to the requirements of 3.3.1 thru 3.3.21 inclusive.

4.8.1 CCA logic. CCA logic shall be demonstrated by:

a. The CCA shall be manually set to Self Test in accordance with 3.3.6.

b. The CCA shall be manually set to READY state in accordance with 3.3.10.

c. /READY indicator output shall be active.

d. The CCA shall perform a measurement.

e. With a display unit conforming to the requirements of 3.3.3 connected to the CCA, it shall display the velocity of 911.5+/-.2.

f. The /STAND BY indicator shall be active.

4.8.2 Connector. Connector shall be inspected for shorts between the Pins of P5 in accordance with TABLE I.

4.8.3 Display driver. Display driver shall be tested by connecting a display unit in accordance with requirements of

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3.3.3.1 and TABLE II, and displaying the following minimum patterns:

a. Test pattern No. 1 : 333.8.
b. Test pattern No. 2 : 911.5.
c. Test pattern No. 3 : 8 8888.8.

4.8.4 DISP SW. DISP SW input function shall be demonstrated by:

a. Setting the CCA measurement rounds table in CCA memory by means of the serial communication to the following combination:

Measurement rounds table	Setting
Current measurement	0000.0
Current - 1	1111.1
Current - 2	2222.2
Current - 3	3333.3
Current - 4	4444.4
Current - 5	5555.5
Current - 6	6666.6
Current - 7	7777.7
Average	8 8888.8

b. Connecting a display unit with the requirements of 3.3.3.

c. Scanning the DISP SW inputs in accordance with 3.3.4 and obtaining the "measurements" in accordance with 4.8.4(a).

4.8.5 Accuracy. Accuracy shall be tested by injecting the following frequencies and obtaining the following measurements on display driver :

Frequency	Measurements
23,438 +/- 2 Hz	333.8 +/- .2
46,875 +/- 4 Hz	665.4 +/- .2
75,000 +/- 6 Hz	1069.5 +/- .2

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4.8.6 Self test. Self Test shall be demonstrated by setting CCA input pins 40,5,38,4 to TTL Logic High, setting the CCA to READY state in accordance with 3.3.10 and obtaining 911.5 +/- .2 measurement at display driver.

4.8.7 Software. Software shall be tested by verifying that the data stored in the firmware device is identical to a master EPROM. The contents of the master EPROM shall be authenticated by using the software as described in Drawing No. 12922145.

4.8.8 V range. Measurement range shall be tested by injecting a doppler signal at Pin 67 of P5 and measuring bandpass filter properties in accordance with 3.3.8, TABLE IV. Selection of bandpass filter shall be done by applying the command to Pins 40,5,38,4 of P5 in accordance with 3.3.8, TABLE IV.

4.8.9 Automatic reset. Automatic Reset shall be tested by obtaining a reading of 0000.0 of DISPLAY DRIVER after power is applied to the CCA.

4.8.10 Press before fire (PBF). The toggling between READY and STAND BY shall be tested by applying the input command in accordance with 3.3.10 and verifying that /READY indicator and /STAND BY indicator are activated each at the time.

4.8.11 CPU reset. The input command CPU RESET shall be tested after the CCA has performed 8 measurements. Pin 44 of P5 than shall be grounded and DISP SW shall be scanned in accordance with TABLE III, verifying that the Display Driver is driving 0000.0 and 0 0000.0 for average display.

4.8.12 Indicator lamps. Indicator lamps shall be tested by connecting 2 lamps of 28V 40mA in accordance with requirements of 3.3.12. /READY lamp shall be illuminated whenever the CCA is in READY state. /STAND BY lamp shall be illuminated whenever the CCA is in STAND BY state.

4.8.13 Antenna control. Antenna Control shall be tested by loading Pin 67 and Pin 30 of P5 with 2 resistor of 680 ohm in series. When the CCA is in READY state, the voltage in the joint of the two resistors shall be 6 +/- .5 V. When CCA is in STAND BY state the voltage in the joint of the two resistors shall be 12 +/- .9 V.

4.8.14 +24V INPUT PWR MON.

a. When 24V is applied to Pin 68 of P5, the CCA shall enter the READY state.

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b. /READY indicator shall be activated and Antenna Control shall be 6 +/- .5 V in accordance with 4.8.13.

c. Voltage to Pin 68 shall be decreased to within 16.4 +/- .5 V and that /READY indicator is off, and antenna control is 12 +/- .9 V in accordance with 4.8.13.

d. CCA shall enter the STAND BY state, /STAND BY indicator shall be activated.

e. Voltage to Pin 68 shall be decreased to 16.4 +/- .5 V and verifying that /STAND BY indicator is off.

4.8.15 Caliber. CALIBER input shall be tested by setting the CCA to Self Test in accordance with 3.3.6, and measuring the time delay between pulses at Pin 45 of P5. The following results are to be verified:

Pin 43 of P5 10	Time Delay
1 1	8.2 +/- .4 msec
1 0	16.4 +/- .4 msec
0 1	32.9 +/- .4 msec
0 0	49.3 +/- .4 msec

4.8.16 Serial communications. Serial communications shall be tested by connecting a serial communications data terminal, RS-232C. SER DATA OUT, Pin 19 of P5 shall be tested by obtaining the following data on the data terminal every .5 sec approximately:

Y F XXXX.X

where Y designates the status of DISP SW:

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Pin of P5				Y
41	7	6	39	
1	1	1	1	0
1	1	1	0	1
1	1	0	1	2
1	1	0	0	3
1	0	1	1	4
1	0	1	0	5
1	0	0	1	6
1	0	0	0	7
0	1	1	1	8

F designates a constant F for Y=0 to 7, and the number of rounds for average calculation if Y=8. XXXX.X designates the measurement being driven by Display Driver in accordance with TABLE III of 3.3.4. SER DATA IN shall be tested by typing a string on the data terminal and obtaining the "measurements" in accordance with 4.8.4.(a).

4.8.17 Test signals. Test Signals shall be tested for existence using SME.

4.8.18 Power requirements. The CCA shall be tested for power consumption in accordance with 3.3.19, TABLE VII.

5. PACKAGING

5.1 Packaging, packing, and marking. Packaging, packing and marking shall be in accordance with MIL-P-14232 and Special Packaging Instruction P12922117. The level of protection shall be specified in the procurement document.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful but is not mandatory.)

6.1 Intended use. The circuit card assembly is intended for use in the M90 Radar Chronograph data unit.

6.2 Acquisition requirements. Procurement data should specify the following :

a. Title, number and date of this specification.
b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1).

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c. Applicable stock number.
d. First Article Sample requirements (see 3.5).
e. Level of preservation and packing (see 5.1).
f. Serialization requirements, if applicable.
g. Certificate of conformance for each lot or shipment of product.

6.3 Consideration of data requirements. The following data requirements should be considered when this specification is applied on a contract. The applicable Data Item Descriptions (DID's) should be reviewed in conjunction with the specific acquisition to ensure that only essential data are requested/provided and that the DID's are tailored to reflect the requirement of the specific acquisition. To ensure correct contractual application of the data requirements, a Contract Data Requirements List (DD Form 1423) must be prepared to obtain the data, except where DOD FAR Supplement 27.475-1 exempts the requirement for a DOD Form 1423.

Reference Paragraph	DID Title	DID No.	Suggested Tailoring
4.6	Quality Engineering Acceptance Inspection Equipment Description Documentation	DI-R-1714	None

6.4 Drawings. Drawings listed in Section 2 of this specification under the heading U.S Army Armament Research, Development and Engineering Center (ARDEC), may also include drawings prepared by, and identified as, U.S. Army Armament, Research and Development Command (ARRADCOM), Frankford Arsenal, Rock Island Arsenal or Picatinny Arsenal drawings. Technical data originally prepared by these activities is now under cognizance of ARDEC.

6.5 Submission of contractor inspection equipment designs for approval. Submit two copies of designs as required: Commander, U.S. Army AMCCOM, ATTN: AMSMC-QAF-I(D), Picatinny Arsenal, NJ 07806-5000. This address will be specified on the Contract Data Requirements List, DD Form 1423 in the contract. Unless otherwise specified, data item DI-R-1714 or as stated in the contract will apply.

6.6 Definitions.

6.6.1 Contractor. Unless otherwise specified, the word Contractor used throughout this specification shall mean any supplier or producer of items or material to the Government. This shall include commercial contractors, sub-contractors, Government

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Owned-Contractor Operated (GOCO), and Government Owned-Government Operated (GOGO) plants.

6.6.2 Technical data package. A technical description of an item adequate for use in procurement. The description defines the required design configuration and assures adequacy of item performance. It consists of all applicable lists, specifications, standards, models, performance requirements, quality assurance provisions and packaging data.

6.7 Government furnished property. The contracting officer should arrange to furnish the property listed in 3.4.6.

6.8 Subject term (key word) listing.

M90 Radar Chronograph (M90)
Circuit Card Assembly (CCA)
Press Before Fire (PBF)

Custodian:
Army-AR

Preparing activity:
Army-AR

(Project 1290-A455)

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INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
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I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-C-70943 (AR)	30 July 1990

3. DOCUMENT TITLE

CIRCUIT CARD ASSEMBLY, MICROPROCESSOR: 12922117

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code) (1) Commercial	(2) AUTOVON
US Army ARDEC Standardization & Specification Office	(201) 724-6674	880-6674
c. ADDRESS (Include Zip Code) ATTN: SMCAR-BAC-S Picatinny Arsenal, NJ 07806-5000	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

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